

Jenish Twayana

Software Engineer (AI/ML)

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SUMMARY

AI/ML Software Engineer with over two years of experience designing, building, and shipping production generative AI systems, including retrieval-augmented generation (RAG) pipelines, conversational agents, and LLM evaluation frameworks. Strong full-stack development background enables end-to-end ownership, from model and pipeline design through to deployment on AWS. Currently supporting a production AI assistant used by tens of thousands of users, and a computer-vision system running on 200+ deployed devices. Focused on applying AI/ML pragmatically to solve real business problems.

TECHNICAL SKILLS

Languages: Python, TypeScript

AI/ML: PyTorch, TensorFlow, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Amazon Bedrock

Frameworks & Libraries: LangChain, React.js, Node.js, Express.js, Django, Tailwind CSS, Flutter

Cloud & DevOps (AWS): EC2, S3, RDS, ECS, ECR, SageMaker, Bedrock, Transcribe

Databases: PostgreSQL

Version Control: Git, GitHub, Bitbucket

PROFESSIONAL EXPERIENCE

Leapfrog Technology | Kathmandu, Nepal

Software Engineer

August 2024 – Present

- Architected and delivered a HIPAA-compliant speech-to-text (STT) and AI-based feature that automatically fills out forms from audio recordings, taking it from concept to a production-ready release.
- Designed and own the evaluation framework and quality-assurance practices used to benchmark LLM applications across projects, giving teams a consistent way to measure output quality before release.
- Built a RAG-based conversational AI assistant that surfaces HR documentation for healthcare providers, then evolved it into an agent that retrieves live application data to answer questions and take actions; now reaches tens of thousands of users in production.
- Built a low-latency RAG system powering a bot that joins Microsoft Teams calls as a digital avatar and answers questions on tech news and competitor insights, then extended it with agentic capabilities to send emails, run web searches, and take further contextual actions.
- Built a production SQL agent that converts natural-language questions into SQL queries against a traffic database and returns the results in plain language.
- Built a vision-based traffic-monitoring solution with on-device model deployment and a live dashboard, now running across 200+ deployed devices in production.
- Created a RAG-based legal chatbot for Q&A on Nepalese law, including a custom document ingestion pipeline (proof of concept).

Software Engineering Fellow

May 2024 – August 2024

- Learned and applied full-stack fundamentals: HTML, CSS, and JavaScript for the frontend; Node.js and Express.js for the backend; PostgreSQL for data storage.
- Built and deployed multiple projects demonstrating solid frontend-backend integration, following industry best practices.

PERSONAL PROJECTS

Samay Baji — Full-Stack Food Ordering Platform

- Built an end-to-end food ordering platform for local Newari cuisine restaurants, covering authentication, menu management, cart, real-time order tracking, and media handling.
- Developed backend services with Node.js and Express.js, and a frontend using HTML, SASS, and TypeScript.
- Integrated PostgreSQL for relational data, MinIO S3 for media storage, and Docker for containerized deployment.

- Deployed the platform on AWS using Amplify (frontend), API Gateway with Lambda (backend), RDS (database), and Cloudflare (DNS).

Monument Recognition — AI-Powered Mobile App

- Built a mobile app for real-time historical monument recognition using on-device object detection and location intelligence.
- Trained and deployed a MobileNetSSD V2 model on-device and a YOLOv7 model on an Azure-hosted server, both on custom datasets, for monument detection and identification.
- Added GPS validation to cross-check detected monuments against the user's real-time location.
- Delivered nearby amenity recommendations (hotels, ATMs, restaurants) and monument details through a PostgreSQL-backed API.
- Built the Android app in Kotlin/Java, with Python-based ML services hosted on Microsoft Azure.

EDUCATION

Bachelor of Engineering, Computer Engineering

Tribhuvan University (Khwopa College of Engineering), Kathmandu, Nepal | *November 2019 – November 2024*

CERTIFICATIONS

- AWS Certified Machine Learning – Associate (MLA-C01), 2025
- Microdegree in AI, Fusemachines AI Fellowship Program, 2024
- AWS Microcredentials — Serverless, MLOps, Agentic AI

AWS Certification badges: skillsprofile.skillbuilder.aws/user/jenish-twayana/certification-badges